Hsiao-Jou (Amy) Hsu

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Website AGU TV video

Research Interests

Physics-informed and explainable AI for Earth observation; satellite-derived shallow-water bathymetry; models that transfer to unseen regions; quality control of heterogeneous reference data; satellite image processing (cloud removal, mosaicking); open-source geospatial tools.

Education

Ph.D., Geodetic Science, The Ohio State University, Columbus, OH 2023 – present
Advisor: Prof. Joachim Moortgat · Candidacy examination passed May 2024 · GPA 3.92/4.00 · Expected graduation: Spring 2028
Dissertation (working title): *Trustworthy Depth Maps from Space: Physics-Informed, Explainable AI for Shallow-Water Bathymetry*

M.S., Geodetic Science, The Ohio State University, Columbus, OH 2024
Awarded after passing the PhD candidacy examination (December 2024)

M.Sc., Civil Engineering (Geoinformatics), National Central University (NCU), Taiwan 2019 – 2020
Thesis: *Bathymetric Mapping in Shallow Water of the South China Sea by ICESat-2 and Sentinel-2* · GPA 4.00/4.00 · Rank 3/90

B.A., Geography (with Teacher Certification), National Taiwan Normal University (NTNU), Taiwan 2013 – 2017
GPA 4.15/4.30 · Rank 1/77
Undergraduate research: crime-hotspot analysis in Taipei City; influence of water, buildings, roads, topography, and vegetation on the urban cool island in Taipei City

Publications

Peer-reviewed journal articles

[J3] **Hsu, H.-J.**, & Moortgat, J. (2026). From local training to large-scale mapping: A comparative assessment of machine learning and deep learning for transferable satellite-derived bathymetry. *Remote Sensing*, 18(11), 1768. https://doi.org/10.3390/rs18111768 · Code: github.com/buckai-observatory/DL_bathy

[J2] **Hsu, H.-J.**, Tseng, K.-H., Tsai, F., Liu, C.-L., Lo, C.-C., & Moortgat, J. (2026). A novel approach to automated cloud removal and seamless multisensor satellite image mosaicking. *Applied Computing and Geosciences*, 31, 100387. https://doi.org/10.1016/j.acags.2026.100387 · Code: GitHub repository

[J1] **Hsu, H.-J.**, Huang, C.-Y., Jasinski, M., Li, Y., Gao, H., Yamanokuchi, T., Wang, C.-G., Chang, T.-M., Ren, H., Kuo, C.-Y., & Tseng, K.-H. (2021). A semi-empirical scheme for bathymetric mapping in shallow water by ICESat-2 and Sentinel-2: A case study in the South China Sea. *ISPRS Journal of Photogrammetry and Remote Sensing*, 178, 1–19. https://doi.org/10.1016/j.isprsjprs.2021.05.012 · **Cited 174 times** (Google Scholar, October 2026)

Under review

[R2] Moortgat, J., Jain, B., Radhakrishnan, A., Roy Chowdhury, S., & **Hsu, H.-J.** (under review). geoai-datacubes: AI-ready multi-mission satellite data cubes for Earth-observation machine learning. *Journal of Open Source Software*. Software: github.com/buckai-observatory/geoai-datacubes · https://doi.org/10.5281/zenodo.20829119

[R1] **Hsu, H.-J.**, & Moortgat, J. (under review). Seeing beneath the surface: A loss function ablation study for depth-aware shallow-water bathymetry. *International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences* (45th EARSeL Symposium, Athens, Greece, 2026).

Peer-reviewed conference paper

[P1] Chowdhury, S. R., Radhakrishnan, A., **Hsu, H.-J.**, Subramoni, H., & Moortgat, J. (2026). From bands to depth: Understanding bathymetry decisions on Sentinel-2. arXiv:2601.12636, doi:10.48550/arXiv.2601.12636. Preprint of the paper in *2026 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)* (pp. 2063–2072), IEEE, doi:10.1109/WACV61042.2026.00205

In preparation

[S1] **Hsu, H.-J.**, & Moortgat, J. Open-source deep-learning pipeline for transferable satellite-derived bathymetry (`DL_bathy`). Software paper for the *Journal of Open Source Software*, to be submitted December 2026.

Conference Presentations

First author and presenter unless noted.

[C11] **Hsu, H.-J.**, & Moortgat, J. Many views, one seafloor: Physics-informed deep learning for transferable bathymetry. *AGU Fall Meeting*, San Francisco, CA, December 2026 (accepted).

[C10] **Hsu, H.-J.**, & Moortgat, J. Seeing beneath the surface: Depth-aware deep learning for shallow-water bathymetry. *45th EARSeL Symposium*, Athens, Greece, September 2026 (oral).

[C9] **Hsu, H.-J.**, & Moortgat, J. Deep learning–based shallow water bathymetry using Sentinel-2 imagery. *AAAS Annual Meeting*, Phoenix, AZ, February 2026.

[C8] **Hsu, H.-J.**, & Moortgat, J. Can less be more? Assessing the potential of sparse ICESat-2 data for accurate coastal bathymetry mapping. *AGU Fall Meeting*, New Orleans, LA, December 2025.

[C7] **Hsu, H.-J.**, & Moortgat, J. A generalizable framework for shallow water bathymetry using machine learning. *IAG Scientific Assembly*, Rimini, Italy, September 2025 (oral).

[C6] **Hsu, H.-J.**, & Moortgat, J. Enhancing shallow water bathymetry using machine learning with ICESat-2, airborne LiDAR, and Sentinel-2 imagery. *EGU General Assembly*, Vienna, Austria, April 2025.

[C5] **Hsu, H.-J.**, & Moortgat, J. Machine learning–enhanced bathymetry using ICESat-2 altimetry and Sentinel-2 imagery. *AAAS Annual Meeting*, Boston, MA, February 2025.

[C4] **Hsu, H.-J.**, Tseng, K.-H., Bevis, M., Durand, M., & Moortgat, J. An automated algorithm for cloud removal and seamless mosaicking of satellite images. *AGU Fall Meeting*, San Francisco, CA, December 2024.

[C3] **Hsu, H.-J.**, Tseng, K.-H., & Shum, C. K. Shallow water navigation chart in multiple island coastal regions derived from ICESat-2 altimetry and Sentinel-2 imagery. *IUGG General Assembly*, Berlin, Germany, July 2023.

[C2] Tseng, K.-H., **Hsu, H.-J.**, & Pan, P.-C. Coastal bathymetry derived from spaceborne LiDAR and optical images. *43rd Asian Conference on Remote Sensing (ACRS)*, 2022 (co-author).

[C1] **Hsu, H.-J.**, & Tseng, K.-H. Bathymetry mapping in Dongsha Island. *38th Conference on Surveying and Geomatics*, Taoyuan, Taiwan, August 2019.

Honors and Awards

Michael Johnson Graduate Student Award, outstanding PhD candidate starting research, School of Earth Sciences, OSU — 2026

AAAS Section E (Geology and Geography) Student Travel Award, awarded twice — 2024, 2025

Friends of Orton Hall Fund Travel Awards (competitive), School of Earth Sciences, OSU; seven international meetings — 2024 – 2026

Geodetic Science Book Award, School of Earth Sciences, OSU — 2024

Honorable Mention, MATLAB/Simulink Call for Papers, Taiwan — 2020

Graduate Seminar Poster Award, Center for Space and Remote Sensing Research (CSRSR), NCU — 2019

Commendation for organizing the Cross-Strait Groundwater and Engineering Geology Symposium, NCU — 2018

Academic Excellence Award (Dean's List, top 3 in department; three times), NTNU — 2013 – 2017

Honorable Mention, Hand-Drawn Maps Competition, NTNU — 2017

Scholarships for excellent students: Hong-Lin Wu Memorial (Taoyuan County Government), Qing Zong Foundation, Chiefsun Foundation, and Lungshan Temple Foundation (four times) — 2015 – 2016

Financial Support

Graduate Research Assistantship, School of Earth Sciences, OSU — 2023 – 2024; 2025 – present

Graduate Teaching Assistantship, School of Earth Sciences, OSU — 2024 – 2025

OSU Presidential Research Excellence Accelerator award, *Tracking Coastal Security Through Geospatial Satellite Data and Novel AI* ($50,000; PI: J. Moortgat, with H. Subramoni, Computer Science and Engineering); funded graduate research on this project and established the collaboration with Computer Science and Engineering — 2025

Travel awards: Friends of Orton Hall Fund (seven, 2024–2026) and AAAS Section E (two, 2024 and 2025); see Honors and Awards — 2024 – 2026

Graduate research and teaching assistantships, Department of Civil Engineering, NCU (during M.Sc. studies) 2019 – 2020

Research Experience

Graduate Research Assistant, School of Earth Sciences and BuckAI Observatory, OSU 2025 – present
Advisor: Prof. Joachim Moortgat

- Developing an explainable, physics-informed neural network that predicts depth together with physically interpretable maps (light attenuation, seafloor reflectance, seafloor visibility, image quality) linked by the Beer–Lambert law.
- Investigating tide and water-level effects across repeated satellite images.
- Built a standardized multi-region training dataset (∼70,000 patches holding 1.7 million 12-band Sentinel-2 images from 2019–2025, ∼11 TB uncompressed; U.S., New Zealand, Taiwan, Australia) and an automated pipeline for imagery download, vertical-datum conversion, and grid alignment; developing IHO S-44–based quality criteria.
- Benchmarked five ML/DL models for cross-region transfer (Pratas Island and Great Barrier Reef → Ashmore and Cartier Reefs) [J3]; co-developed open-source tools with Computer Science and Engineering [R2, P1].

Graduate Research Assistant, Division of Geodetic Science, OSU 2023 – 2024

Research Assistant, Satellite Receiving Station, CSRSR, NCU (PI: Prof. Fuan Tsai) 2020 – 2022
Projects: *FORMOSAT Series Value-Added Application Data Cube and Environmental Parameters* (2021–2022); *Testing of DSM/DEM Environmental Parameters* (2020)

- Developed an automated cloud-removal and seamless mosaicking workflow for FORMOSAT-2/5, SPOT 6/7, and Pléiades imagery within the FORMOSAT Value-Added Data Cube project; completed at OSU [J2], with cloud-removal rates of 90–98% for SPOT and FORMOSAT-2.
- Tested the transfer of DSM/DEM parameters from SPOT to Pléiades imagery.

Graduate Research Assistant, Department of Civil Engineering, NCU (Advisor: Prof. Kuo-Hsin Tseng) 2019 – 2020
Projects: *Bathymetry Retrieving in the South China Sea*; *Using Multitemporal Satellite Images to Monitor Coastal Changes in Taoyuan*

- Combined ICESat-2 photon-counting lidar with Sentinel-2 imagery to map South China Sea reef bathymetry [J1]; developed photon-denoising and bathymetric-extraction algorithms for the coral-reef islands.
- Monitored coastal change in Taoyuan with multi-temporal satellite imagery.

Research Assistant, Graduate Institute of Applied Geology, NCU (PI: Prof. Chyi-Tyi Lee) 2018 – 2019
Project: *Landslide Sensitive Area Status Analysis and Digitizing* (2018): landslide-susceptibility analysis and satellite-image digitizing.

Intern, GOYOURLIFE Inc., Taiwan 2016

Mentoring and Collaboration

Most senior PhD student in Prof. Moortgat's group and first PhD student of the BuckAI Observatory; mentor and collaborator for graduate students across disciplines 2025 – present

- Computer Science and Engineering graduate students Bhavika Jain, Satyaki Roy Chowdhury, and Aswathnarayan Radhakrishnan: geoai-datacubes [R2], explainable AI for bathymetry [P1], and a PlanetScope satellite-imagery project in Uganda.
- Dylan M. Smith: virtual-reality applications; Samir Pahari: first-year PhD student (Earth-observation data cubes and flood mapping).

Teaching

Graduate Teaching Assistant, OSU: EEOB 2911 *The Climate Crisis* (Autumn 2024); Earth Science Lab (Spring and Summer 2025) 2024 – 2025

Lecturer, QGIS change-detection workshop for senior high-school teachers, NCU 2021 – 2022

Teaching Assistant, Surveying Practice, Department of Civil Engineering, NCU 2019 – 2020

Intern Teacher, Geography, Taoyuan Municipal Nei-Li Senior High School, Taiwan 2017 – 2018

Student Teacher, Geography, Taipei Municipal HuaJiang High School and Jinhua Junior High School 2016 – 2017

Outreach, Media, and Service

Featured in AGU TV's Thought Leadership Film Series (AI in Earth and environmental sciences): 2026
youtube.com/watch?v=BqAkyLukwXI

Treasurer, International Friendships (registered Ohio State student organization serving international students and scholars): manage budgets, expenses, reimbursements, and funding allocations in line with university policies; help plan community events (welcome party, ice cream social, Lunar New Year celebration) that connect students with one another — 2026 – present

Co-organizer, Ecosystem Ecology Workshop for high-school teachers (new curriculum guidelines), Nei-Li Senior High School — 2017 – 2018

Organizing committee, Cross-Strait Groundwater and Engineering Geology Symposium, NCU — 2018

Volunteer, International Student Volunteer Group, NTNU — 2013 – 2017

Affiliations and Memberships

BuckAI Observatory, OSU (new center of excellence in applied AI): first PhD student affiliated with the center — 2025 – present

American Geophysical Union (AGU) — 2024 – present

American Association for the Advancement of Science (AAAS) — 2025 – present

European Geosciences Union (EGU) — 2025

Skills

Programming	Python (PyTorch, rasterio, GDAL, xarray, Zarr), MATLAB; Slurm/HPC, Docker, Git
Machine learning	CNNs (U-Net, DeepLabV3+, ResNet, EfficientNet, ConvNeXt), random forests, physics-informed and explainable deep learning
Remote sensing	Sentinel-2, Landsat, SPOT 6/7, FORMOSAT-2/5, Pléiades, ICESat-2 ATLAS, airborne lidar; Google Earth Engine, SNAP, QGIS, ArcGIS, ERDAS Imagine, GMT
Languages	Mandarin Chinese (native), English (fluent)